

# FOR SERVICE TECHNICIAN'S USE ONLY

NOTE: This sheet contains important Technical Service Data.

W10867183B

Tech Sheet

Do Not Remove or Destroy

## ⚠ DANGER



### Electrical Shock Hazard

Only authorized technicians should perform diagnostic voltage measurements.

After performing voltage measurements, disconnect power before servicing.

Failure to follow these instructions can result in death or electrical shock.

## ⚠ WARNING



### Electrical Shock Hazard

Disconnect power before servicing.

Replace all parts and panels before operating.

Failure to do so can result in death or electrical shock.

## Voltage Measurement Safety Information

When performing live voltage measurements, you must do the following:

- Verify the controls are in the off position so that the appliance does not start when energized.
- Allow enough space to perform the voltage measurements without obstructions.
- Keep other people a safe distance away from the appliance to prevent potential injury.
- Always use the proper testing equipment.
- After voltage measurements, always disconnect power before servicing.

## SPECIFICATIONS

### ELECTRICAL SUPPLY:

(Under load) 60 Hz, 120 VAC

### SUPPLY WATER FLOW RATE:

To fill 2 qt (1.9 L) in 27 seconds, 120 psi maximum, 20 psi minimum.

### SUPPLY WATER TEMPERATURE:

120°F (49°C) (Before starting a cycle, run water from sink faucet until hot.)

### WATER CHARGE:

1.58 gal (6.0 L) approximate

### LOWER SPRAY ARM ROTATION:

12 to 40 rpm

### UPPER SPRAY ARM ROTATION:

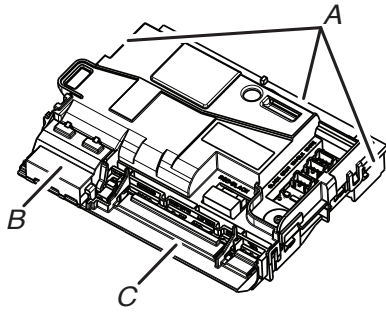
12 to 30 rpm

### REPAIR KITS

Vinyl Touch-Up Kits:

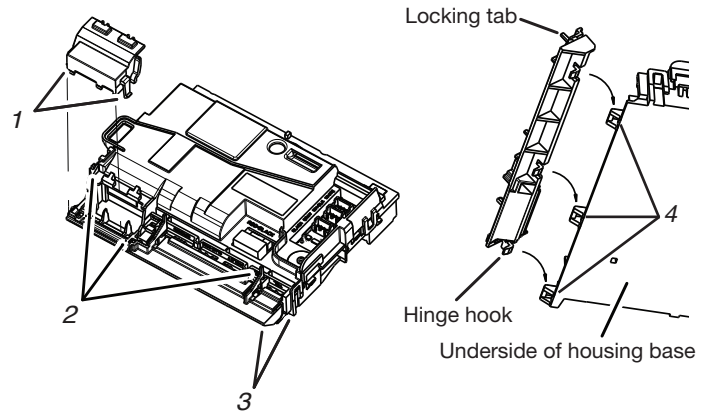
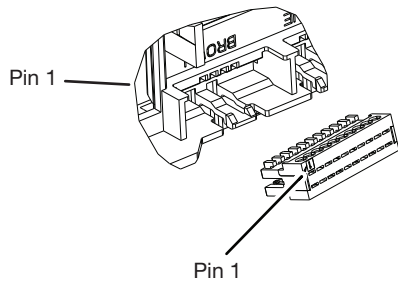
675576 (Blue), 676453 (White), 676455 (Gray)

## Control Assembly



- A.** Control is mounted to the console by snaps along the side and top, and/or screw tabs along the top.
  - B.** Connector Box – Covers high-voltage connections. Must be removed first, then remove connector brace to service control and wiring harness.
  - C.** Connector Brace – Locks wiring harness connectors to control.
- IMPORTANT:** Connector box and brace must be reattached to control when service is completed.

Rast Connector Pinout



### Steps to Access Control for Servicing

- 1. Connector Box Removal:**  
Press inward on the retainer tabs as indicated while lifting up on the connector box. This can be done with a screwdriver or similar tool from either the top or bottom side of the control.
- 2. Connector Brace Removal:**  
Locate the 3 brace locking tabs, as shown. Push each tab in the direction required while applying light downward rotational force on the brace. It is easiest to unlock one tab at a time, working from one end of the brace to the other.
- 3.** Once all 3 tabs are unlocked, pivot the brace to allow removal and gain access to the wiring harness connectors and control for required servicing.
- 4. Connector Brace Reattachment:**  
Attach the brace hinge hooks into holes in the hinge tabs on control housing base. Pivot brace until all 3 locking tabs snap into housing lid. Make sure the harness wires lay flat and are not pinched.
- 5. Connector Box Reattachment:**  
Position the connector box over P4 and P5 connectors, insert the retainer tabs into brace mounting holes, and snap box onto control housing lid. Make sure box is sitting flush against both the brace and housing and harness wires are flat and not pinched.

## Meter Check of Loads and Fuses

### Fuse Service Check:

F601 = Small triac load fuse  
F600 = Wash pump motor fuse  
Check operation of loads during Service Diagnostics cycle.

- If any of the triac loads work (fill, dispenser, drain), F601 fuse is OK.
- If all triac loads fail to work, F601 fuse could be open.
- If wash pump motor runs, F600 is OK.
- If wash pump motor does not run, F600 fuse could be open. If either fuse could be open, See "Fuse Resistance Check."

### Fuse Resistance Check:

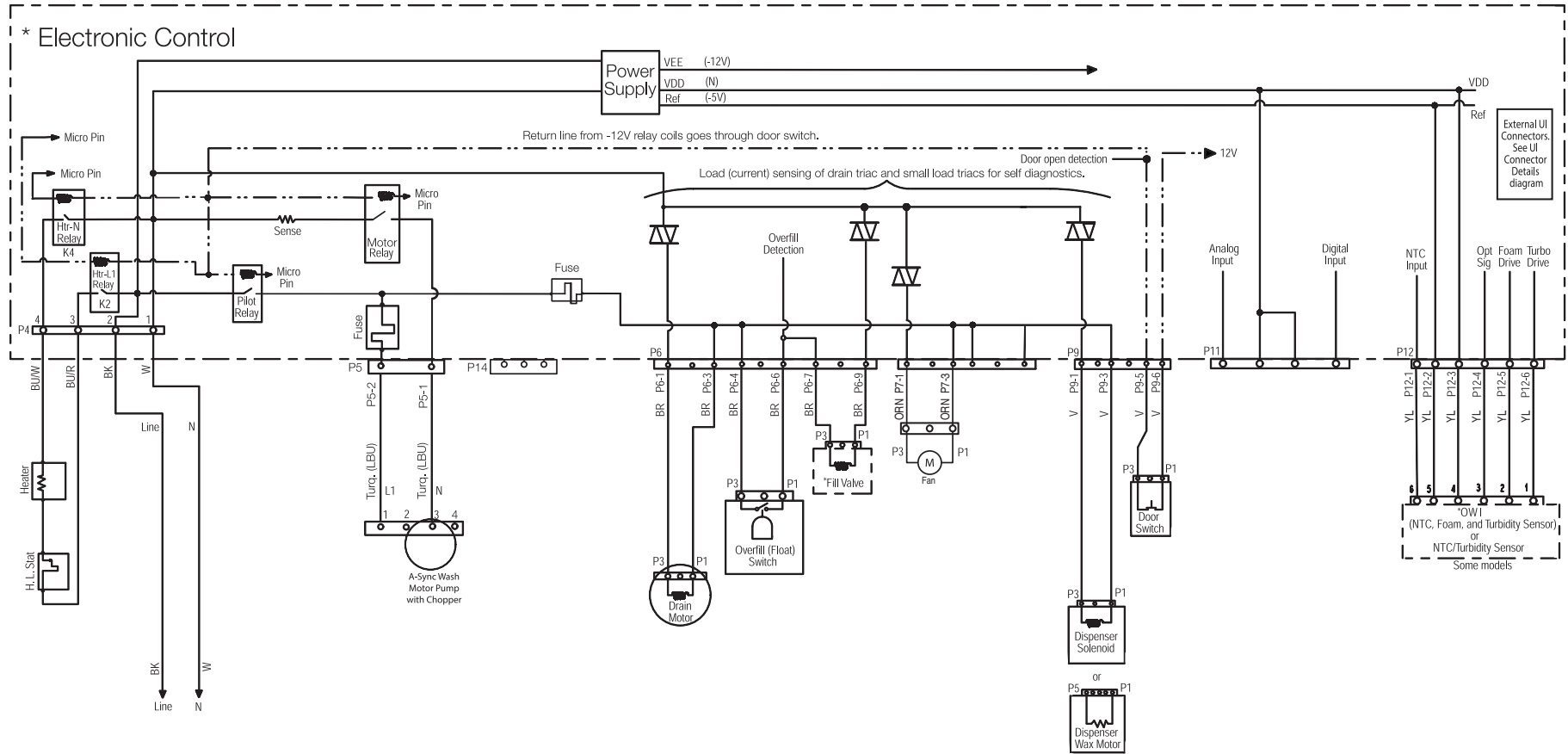
Unplug dishwasher or disconnect power.  
Measure resistance of fuses F600 and F601.

- a.** Disconnect wash pump motor wiring jumper P5 from the control board.
  - b.** Using a resistance meter, measure the resistance between control board connection pins P5-2 and P6-3. See "Control Pinout" diagram.
- If resistance is  $<3 \Omega$ , both fuses are OK.
  - If resistance is  $>3 \Omega$ , replace control.

### What To Do If Fuse Open:

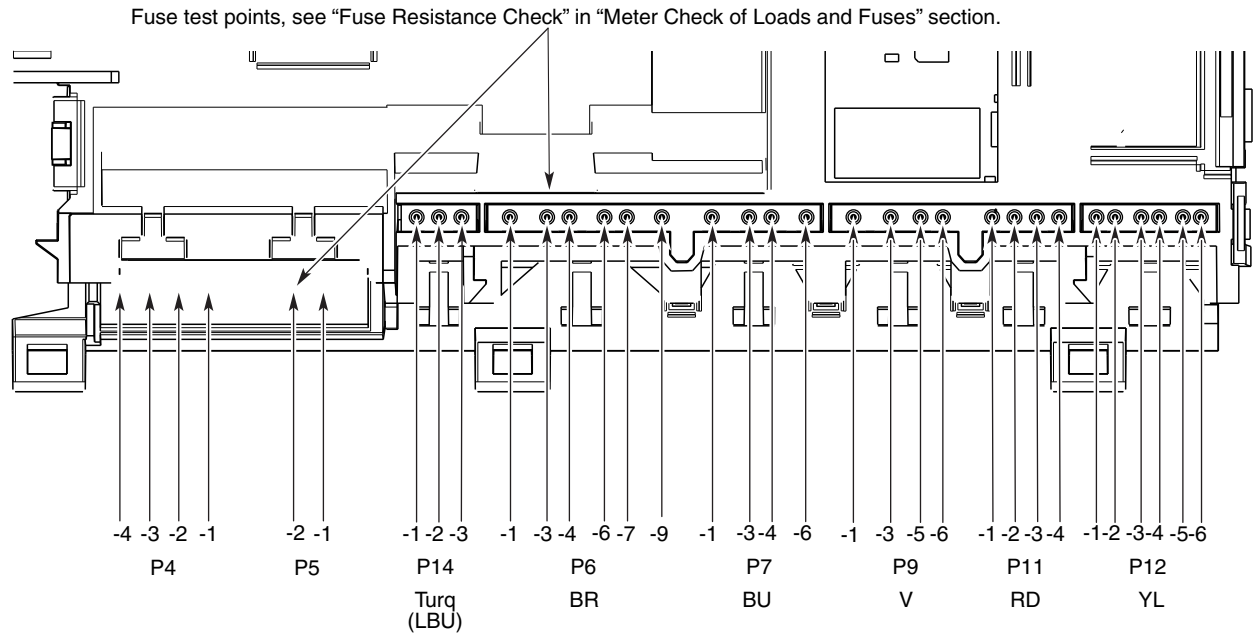
Inspect and check resistance of all loads on fuses. If any loads are open, shorted, or have evidence of overheating or pinched wires, replace loads and/or repair wires.

Schematic shown with door switch and all other normally open contacts open.  
 \*Denotes energy efficient components. Do not substitute.



Metering of Triac Loads

Load must be connected for triac to operate correctly. Meter checks are best made at the control. See illustration below and “Dishwasher Strip Circuits.”

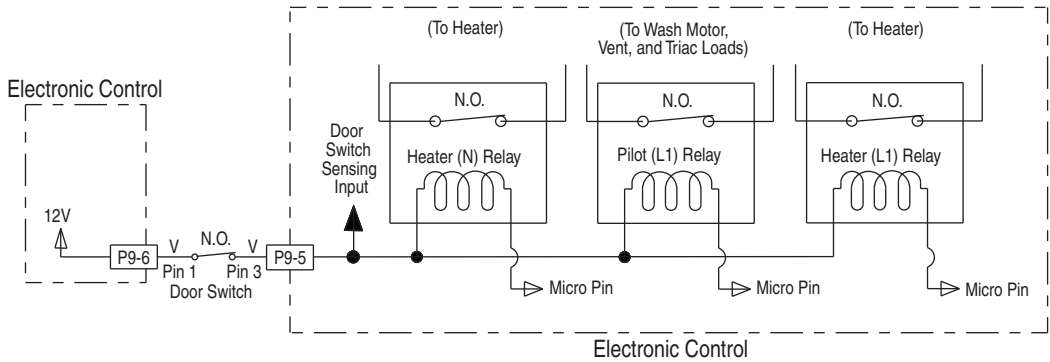


Dishwasher Strip Circuits

The following individual circuits are for use in diagnoses. Do not continue with the diagnosis of the appliance if a fuse has blown, a circuit breaker is tripped, or if there is less than a 120-volt power supply at the wall outlet.

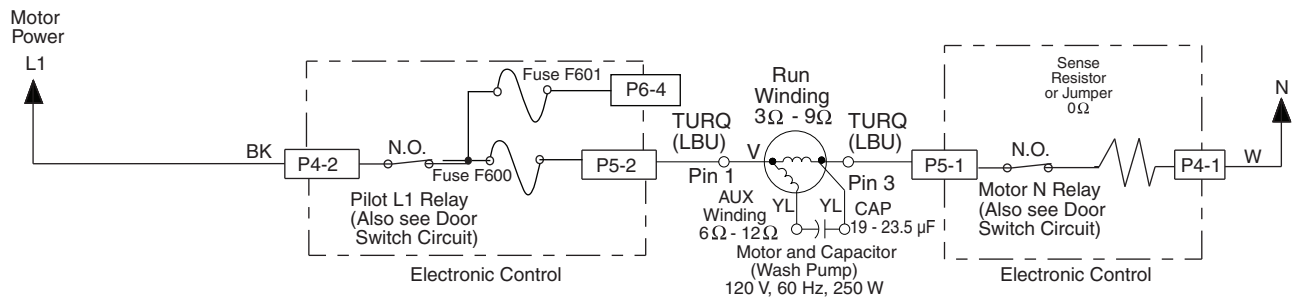
- Unplug dishwasher or disconnect power.
- Perform resistance checks. To check resistance of a component, disconnect harness leads first.

Door Switch



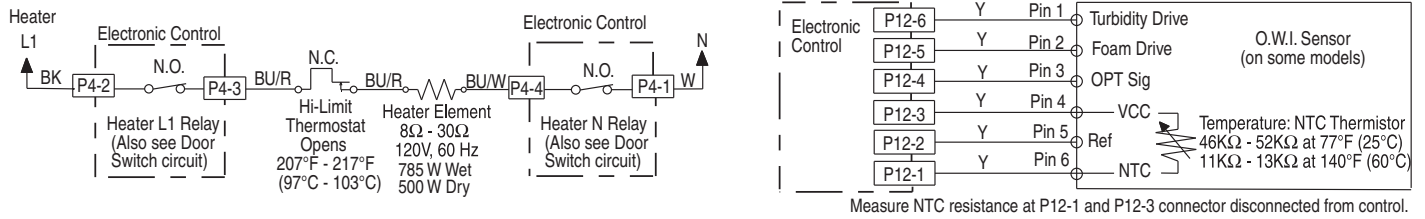
Wash/Rinse

See “Meter Check of Loads and Fuses” to diagnose possible open fuse issues.



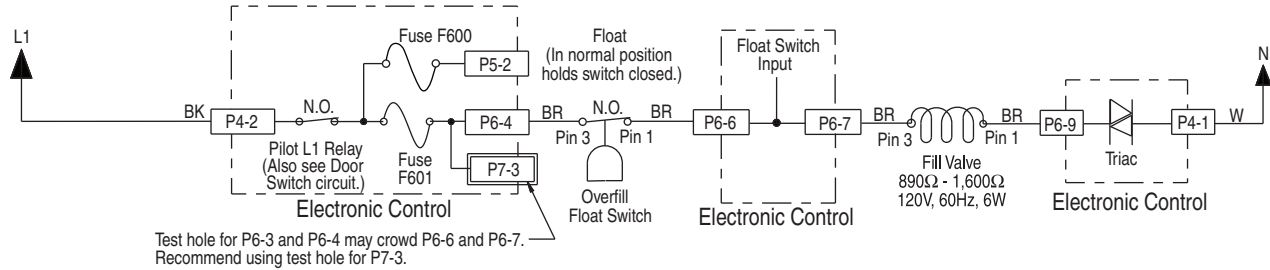
## Water Heating/Heat Dry and Water Sensing with O.W.I. Sensor (Water/Air/Soil/Temperature)

Pump is washing and control monitors temperature during water heating periods. See “Wash/Rinse” and “Water Sensing with O.W.I. Sensor (Water/Air/Soil/Temperature)” circuits.



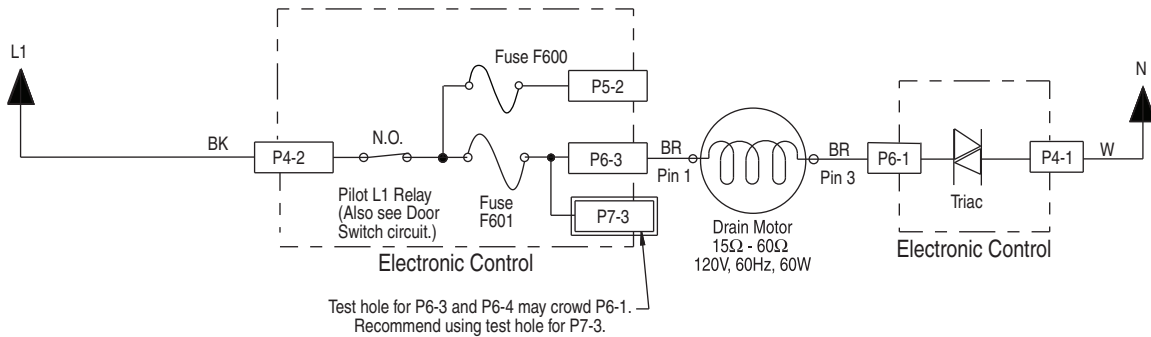
## Fill

See “Meter Check of Loads and Fuses” to diagnose possible open fuse issues.



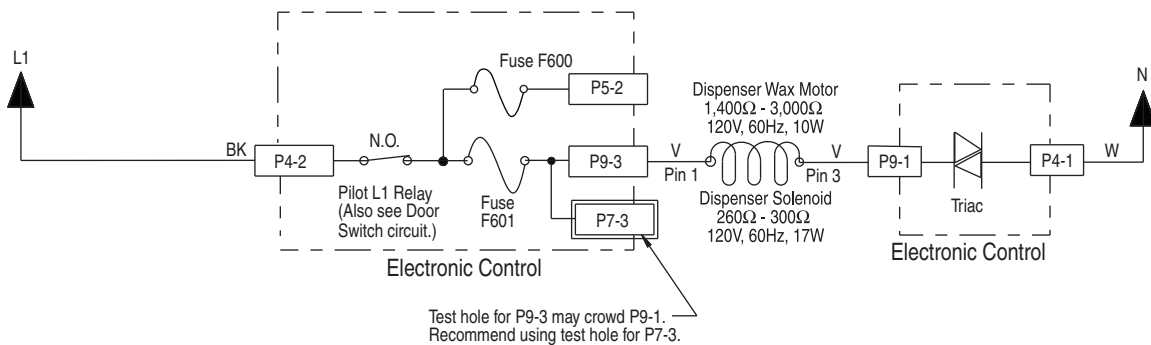
## Drain

See “Meter Check of Loads and Fuses” to diagnose possible open fuse issues.



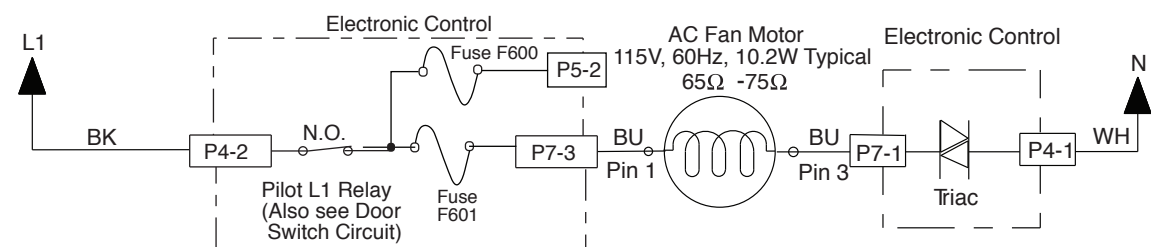
## Dispenser (Detergent and Rinse Aid)

See “Meter Check of Loads and Fuses” to diagnose possible open fuse issues.



## AC Fan

See “Meter Check of Loads and Fuses” to diagnose possible open fuse issues.



| INTERVAL                                      | 33   | 32   | 31               | 30               | 29               | 28               | 27   | 26   | 25   | 24   | 23   | 22   | 21   | 20    | 19   | 18   | 17    | 16   | 15   | 14   | 13    | 12   | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4    | 3     | 2     | 1    |     |  |  |  |
|-----------------------------------------------|------|------|------------------|------------------|------------------|------------------|------|------|------|------|------|------|------|-------|------|------|-------|------|------|------|-------|------|------|------|------|------|------|------|------|------|-------|-------|------|-----|--|--|--|
| CYCLE, OPTION, AND STATUS LEDs                |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| Cycle 1                                       | CY1  |      |                  | CY1              |                  | CY1              | CY1  |      | CY1  |      | CY1  |      | CY1  |       | CY1  |      | CY1   |      | CY1  |      | CY1   |      | CY1  |      | CY1  |      | CY1  |      | CY1  |      | CY1   |       | CY1  |     |  |  |  |
| Cycle 2                                       | CY2  |      | CY2              |                  | CY2              |                  |      | CY2  |      | CY2  |      | CY2  |      | CY2   |      | CY2  |       | CY2  |      | CY2  |       | CY2  |      | CY2  |      | CY2  |      | CY2  |      | CY2  |       | CY2   |      | CY2 |  |  |  |
| Heated Dry                                    | HD   |      |                  | HD               |                  | HD               |      |      | HD   |      | HD   |      | HD   |       | HD   |      | HD    |      | HD   |      | HD    |      | HD   |      | HD   |      | HD   |      | HD   |      | HD    |       | HD   |     |  |  |  |
| Hi Temp                                       | HIT  |      | HIT              |                  | HIT              |                  | HIT  | HIT  |      | HIT  |      | HIT  |      | HIT   |      | HIT  |       | HIT  |      | HIT  |       | HIT  |      | HIT  |      | HIT  |      | HIT  |      | HIT  |       | HIT   |      | HIT |  |  |  |
| Start/Resume                                  | STA  |      | STA              | STA              | STA              | STA              | STA  | STA  | STA  | STA  | STA  | STA  | STA  | STA   | STA  | STA  | STA   | STA  | STA  | STA  | STA   | STA  | STA  | STA  | STA  | STA  | STA  | STA  | STA  | STA  | STA   | STA   | STA  | STA |  |  |  |
| Sanitized<br>NOTE 4                           | SAN  |      |                  |                  |                  |                  |      |      |      |      |      |      |      | (SAN) |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| Clean/Complete - 7 seg<br>NOTES 3, 4, 5       | CLN  |      | 7seg             | 7seg             | 7seg             | 7seg             |      |      |      |      |      |      |      | (CLN) |      |      | (CLN) |      |      |      | (CLN) |      |      |      |      |      |      |      |      |      |       | 7seg  | 7seg |     |  |  |  |
| All other Cycle, Option, and Status LEDs      | ALL  |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| INTERNAL TIME (min:sec)                       | 0:05 | 0:01 | 0:08             | 0:08             | 0:08             | 0:08             | 0:10 | 0:01 | 0:45 | 0:13 | 0:05 | 0:10 | 0:03 | 1:00  | 0:05 | 0:30 | 0:30  | 0:05 | 2:00 | 0:15 | 1:00  | 1:00 | 0:01 | 0:01 | 0:01 | 0:01 | 0:15 | 1:40 | 0:01 | 0:05 | 1:55  | 0:08  | 0:08 |     |  |  |  |
| TOTAL TIME (MAX): 23:24                       |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| (Temperature Sensor)<br>NOTE 4                |      |      |                  |                  |                  |                  |      |      |      |      | THR  | THR  | THR  |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| (Optical Soil Sensor)<br>NOTE 5               |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      | OWI  |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| NOTE: OWI has thermistor built in (see above) |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| (Virtual Water Indicator)<br>NOTE 8           |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      | VWI  |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| LOADS                                         |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| Pilot Relay                                   |      |      | Customer Error 1 | Customer Error 2 | Customer Error 3 | Customer Error 4 |      |      | PLT  | PLT  | PLT  | PLT  | PLT  | PLT   | PLT  | PLT  | PLT   | PLT  | PLT  | PLT  | PLT   | PLT  | PLT  | PLT  | PLT  | PLT  | PLT  | PLT  | PLT  | PLT  | PLT   | PLT   |      |     |  |  |  |
| Fill Valve                                    |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| Wash Pump Motor                               |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| Dispenser (detergent/rinse aid)               |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| Drain Pump Motor                              |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| Heater                                        |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      |       |       |      |     |  |  |  |
| AC Fan (not all models)                       |      |      |                  |                  |                  |                  |      |      |      |      |      |      |      |       |      |      |       |      |      |      |       |      |      |      |      |      |      |      |      |      | (FAN) | (FAN) |      |     |  |  |  |
| REFERENCE NOTES (see below)                   |      |      | 1                | 1                | 1                | 1                | 2    |      |      |      |      |      |      | 3     |      |      | 5     |      |      |      | 4     |      |      |      |      |      |      |      |      | 6    | 6     |       |      |     |  |  |  |

## Service Diagnostics Cycle Notes

- To invoke the Service Diagnostics cycle, perform the following while in Standby:
  - Press any 3 keys in the sequence 1-2-3-1-2-3-1-2-3 with no more than 1 second between key presses.
  - The Service Diagnostics cycle will start when the door is closed.
  - To rapid advance 1 interval at a time, press START/RESUME. Rapid advance may skip sensor checks as some checks require 2 complete intervals.

**NOTE:** The Service Diagnostic cycle will pause when the door is opened and resume automatically upon door closure. No Start/Resume key press is required to resume.

  - Invoking Service Diagnostics cycle clears all status and last run information from memory and restores defaults. It also forces the next cycle to be a sensor calibration cycle. Calibration cycle may add additional rinses prior to the final rinse to ensure clear water and then calibrates the OWI during the fill at the beginning of the final rinse.
  - Drain and wash motors will pulsate on and off.
  - Last Ran cycles and options returned to default.
  - Last Ran Delay returns to the default delay setting.
  - Operating state returns to Standby upon completing or terminating the Service Diagnostics cycle.

- Press Hi Temp or Cycles key in this interval to clear customer error history.
- Thermistor (temperature sensor) checks:
  - Turn Clean LED on if thermistor is in its normal temperature range of 32°F–167°F (0°C–75°C).
  - Turn Sanitized LED on if Fill temperature is above 156°F (69°C).
- OWI (Optical Soil Sensor) Checks:
  - Turn on CLEAN/COMPLETE LED if elevated soil levels are detected during the previous OWI sensing interval.
- Turn on Clean LED in this interval if Stable water condition is detected during the previous VWI sensing interval.
- Not all models have a Fan.

## Customer Cycle Operation

To quickly advance through customer cycles, invoke the Rapid Advance mode by pressing HIGH TEMP - HEATED DRY - HIGH TEMP - HEATED DRY, after starting the cycle. Then, press START/RESUME to advance through cycle intervals.

**NOTE:** Rapid Advance mode is automatically enabled in the Service Diagnostic cycle but must be manually invoked in customer cycles.

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## Service Diagnostics with Error Codes

Entry sequence: Press any 3 keys in the sequence 1-2-3-1-2-3-1-2-3 with no more than 1 second between key presses.

|                                                                                                                                                                                              |                |                                       |               |                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------|---------------|------------------------------------------------------|
| DISPLAY TEST - ALL LEDs ON                                                                                                                                                                   |                |                                       | INTERVAL 33   |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| DISPLAY TEST - ALL LEDs OFF                                                                                                                                                                  |                |                                       | INTERVAL 32   |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| ERROR – MOST RECENT                                                                                                                                                                          |                |                                       | INTERVAL 31   |                                                      |
| 7-SEG will display FUNCTION code (F#).                                                                                                                                                       | Pause 0.5 sec. | 7-SEG will display PROBLEM code (E#). | Pause 1 sec.  | Repeat 3 times unless advanced by Start/ Resume key. |
| If no error, 7-SEG will display (F-).                                                                                                                                                        |                | If no error, 7-SEG will display (E-). |               |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| ERROR 2                                                                                                                                                                                      |                |                                       | INTERVAL 30   |                                                      |
| 7-SEG will display FUNCTION code (F#).                                                                                                                                                       | Pause 0.5 sec. | 7-SEG will display PROBLEM code (E#). | Pause 1 sec.  | Repeat 3 times unless advanced by Start/ Resume key. |
| If no error, 7-SEG will display (F-).                                                                                                                                                        |                | If no error, 7-SEG will display (E-). |               |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| ERROR 3                                                                                                                                                                                      |                |                                       | INTERVAL 29   |                                                      |
| 7-SEG will display FUNCTION code (F#).                                                                                                                                                       | Pause 0.5 sec. | 7-SEG will display PROBLEM code (E#). | Pause 1 sec.  | Repeat 3 times unless advanced by Start/ Resume key. |
| If no error, 7-SEG will display (F-).                                                                                                                                                        |                | If no error, 7-SEG will display (E-). |               |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| ERROR 4 – OLDEST                                                                                                                                                                             |                |                                       | INTERVAL 28   |                                                      |
| 7-SEG will display FUNCTION code (F#).                                                                                                                                                       | Pause 0.5 sec. | 7-SEG will display PROBLEM code (E#). | Pause 1 sec.  | Repeat 3 times unless advanced by Start/ Resume key. |
| If no error, 7-SEG will display (F-).                                                                                                                                                        |                | If no error, 7-SEG will display (E-). |               |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| 10-second pause                                                                                                                                                                              |                | Hi Temp LED will be on.               | INTERVAL 27   |                                                      |
| Press HI TEMP to clear errors.                                                                                                                                                               |                |                                       |               |                                                      |
| Hi Temp LED will blink twice to indicate errors have been cleared. If the Hi Temp key does not respond, open and close the door to wake up the control panel. Press HI TEMP to clear errors. |                |                                       |               |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| Service Diagnostics Cycle<br>Turns on loads and checks sensors.                                                                                                                              |                |                                       | INTERVAL 26-3 |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| SERVICE CYCLE ERROR 1                                                                                                                                                                        |                |                                       | INTERVAL 2    |                                                      |
| 7-SEG will display FUNCTION code (F#).                                                                                                                                                       | Pause 0.5 sec. | 7-SEG will display PROBLEM code (E#). | Pause 1 sec.  | Repeat 3 times unless advanced by Start/ Resume key. |
| If no error, 7-SEG will display (F-).                                                                                                                                                        |                | If no error, 7-SEG will display (E-). |               |                                                      |
| ↓                                                                                                                                                                                            |                |                                       |               |                                                      |
| SERVICE CYCLE ERROR 2                                                                                                                                                                        |                |                                       | INTERVAL 1    |                                                      |
| 7-SEG will display FUNCTION code (F#).                                                                                                                                                       | Pause 0.5 sec. | 7-SEG will display PROBLEM code (E#). | Pause 1 sec.  | Repeat 3 times unless advanced by Start/ Resume key. |
| If no error, 7-SEG will display (F-).                                                                                                                                                        |                | If no error, 7-SEG will display (E-). |               |                                                      |

**NOTE:** Once error codes are extracted, refer to the “Service Error Codes Table” section to diagnose and correctly resolve the root cause condition.

## Service Error Codes Table

Example: 6-1 means “Inlet Water” function, “Low/No Water” problem.

| Function Code     | Problem Code              | Causes                                                                                                                                                                                                    | What to Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1- Control        | 1- Pilot Stuck On         | Control detected K1 pilot relay stuck closed.                                                                                                                                                             | 1. Unplug dishwasher or disconnect power.<br>2. Check all loads on K1 Pilot Relay for shorts.<br>3. Replace control and all shorted components.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                   | 2- Control Software Issue | All LEDs are on.<br><br>Damaged or corrupted memory on control board. Incompatible software components inside micro.                                                                                      | 1. Unplug dishwasher or disconnect power.<br>2. Replace UI. (for models with numeric display only).<br><br>1. Unplug dishwasher or disconnect power.<br>2. Replace control board.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2- User Interface | 1- Stuck Key              | Control detected stuck key(s) in keypad or keypad connection.<br><b>NOTE:</b> If any keys are stuck, the stuck key will be ignored and an error recorded to the service history but no alert to customer. | Check responsiveness of each key.<br>1. If some keys do not respond:<br>- Unplug dishwasher or disconnect power.<br>- Disassemble door and disconnect keypad connection from control or LCD display module.<br>- Verify all other connections to control are made.<br>- Reassemble door, but do not close door (leave keypad disconnected).<br>- Plug in dishwasher or reconnect power.<br>- Wait at least 7 seconds for control to power up completely.<br>- Close dishwasher door and monitor control response:<br><b>A.</b> If control is OK (no longer sees stuck keys with keypad unplugged), it will respond by turning on the drain motor for 2 minutes. Replace keypad and console.<br><b>B.</b> If control is not OK (still sees stuck keys with keypad unplugged), it will not turn on drain motor. Wait for at least 10 seconds. If still no drain response, then replace control or LCD display module (whichever one the keypad was connected to).<br><b>C.</b> If all keys appear OK or intermittent, and keypad is capacitive touch type, then:<br>- Verify tub brackets are screwed to underside of countertop and not hanging over keys (if screw head too close, relocate screw to alternate hole).<br>- Check for evidence of moisture or debris on the surface of the keys. If evident, clean and instruct customer about keeping surface clean.<br>- Check error code history for Fan Error 10-3 as potential cause of condensation on user interface. |



| Function Code             | Problem Code           | Causes                                                                                                                                                                                    | What to Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2- User Interface (cont.) | 2- No response from UI | User interface cannot communicate with main control. Loose user interface connection.                                                                                                     | <ol style="list-style-type: none"> <li>1. Unplug dishwasher or disconnect power.</li> <li>2. Check the connection between the UI and the control P1A connector. If the connection(s) are loose or if wires are loose or damaged; reconnect, repair, or replace as needed.</li> <li>3. Check for 5 VDC from P1A-2 to P1A-4. If there is no voltage at control; disconnect power, replace main control. If there is voltage at control; disconnect power and replace console.</li> </ol> |
| 3- Ther-mistor/ OWI       | 1- Open                | Open connector or component in Temperature Sensing circuit. Open or faulty temperature sensor. Temperature sensor input on control.                                                       | <ol style="list-style-type: none"> <li>1. Check operation of temperature sensor in Service Diagnostics cycle.</li> <li>2. Unplug dishwasher or disconnect power.</li> <li>3. Check all components and connections in the Temperature Sensing circuit with meter. Fix/ replace open connection /part.</li> </ol>                                                                                                                                                                        |
|                           | 2- Shorted             | Incoming water temperature above 167°F (75°C). Shorted connection or component in Temperature Sensing circuit. Shorted or faulty temperature sensor. Temperature sensor input on control. | <ol style="list-style-type: none"> <li>1. Check incoming water temperature.</li> <li>2. Check operation of temperature sensor in Service Diagnostics cycle.</li> <li>3. Unplug dishwasher or disconnect power.</li> <li>4. Check all components and connections in the Temperature Sensing circuit with meter. Fix/ replace open connection /part.</li> </ol>                                                                                                                          |
|                           | 3- Failed Calibration  | 1. OWI failure                                                                                                                                                                            | <ol style="list-style-type: none"> <li>1. Check all connections in Soil Sensing circuit.</li> <li>2. Check OWI lens surface. Clean if needed.</li> <li>3. Run Service Diagnostics to check OWI operation.</li> </ol> <p>OWI should see low soil with clear water. Unplug dishwasher or disconnect power and replace OWI or control if needed.</p> <p>NOTE: Run diagnostics after installing OWI to force calibration on next regular wash cycle.</p>                                   |
|                           |                        | 2. Drain hose check valve not sealing.                                                                                                                                                    | <p>Dirty water backs into dishwasher after draining.</p> <ol style="list-style-type: none"> <li>1. Disconnect drain hose at plumbing connection.</li> <li>2. Elevate hose above dishwasher and fill with water. If water flows into dishwasher, replace entire drain loop (install as high as possible and attach to underside of countertop if possible).</li> </ol>                                                                                                                  |
| 4- Wash Motor             | 3- Motor Not Running   | 1. Loose connection in Motor circuit and/or faulty wash motor.                                                                                                                            | <ol style="list-style-type: none"> <li>1. Check operation of wash motor during diagnostics.</li> <li>2. Unplug dishwasher or disconnect power.</li> <li>3. Check resistances of connections in the Wash circuit.</li> </ol> <p>- Check/fix loose connections or replace wash motor.</p>                                                                                                                                                                                                |
|                           |                        | 2. Control Motor Drive circuit or Sense circuit.                                                                                                                                          | <ol style="list-style-type: none"> <li>1. Unplug dishwasher or disconnect power.</li> <li>2. If meter check of Wash Motor circuit shows normal resistance and still not getting power to the wash motor, replace control.</li> </ol>                                                                                                                                                                                                                                                   |

| Function Code  | Problem Code                         | Causes                                                                                                                                                                                                                                                                                      | What to Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5- Door Switch | 1- Door Stuck Open                   | 1. Door was not latched within 4 seconds of pressing the Start/Resume key.                                                                                                                                                                                                                  | Instruct customer. Refer to Use and Care Guide.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                |                                      | 2. Loose connection in Door Switch circuit and/or door switch contacts stuck open and/or door switch not making contact:<br>- Sloppy door latch assembly (which can be aggravated by high door closure force, keeping strike plate from fully seating).<br>- Door switch (high resistance). | <ol style="list-style-type: none"> <li>1. Check strike plate and door closure force. Verify door seal is seated properly. Check for interference between dish racks and door. Try bending strike plate down for better engagement.</li> <li>2. Unplug dishwasher or disconnect power.</li> <li>3. Check resistances of door switch contacts and all connections in the Door Switch circuit while opening and closing the door latch.</li> <li>- If high resistance with door closed, check/fix loose connections.</li> <li>4. Measure resistance of door switch contacts while checking mechanical operation of latch assembly. Check for broken plastic pieces on latch assembly. Replace latch if faulty.</li> </ol> |
|                |                                      | 3. If none of the above.                                                                                                                                                                                                                                                                    | <ol style="list-style-type: none"> <li>1. Unplug dishwasher or disconnect power.</li> <li>2. Check for 13 VDC from P9-5 to P9-6 by completing the following steps:<br/><b>A.</b> Connect voltage measurement equipment.<br/><b>B.</b> Plug in dishwasher or reconnect power and verify 13 VDC.<br/><b>C.</b> If no voltage is present, unplug dishwasher or disconnect power and replace control.</li> </ol>                                                                                                                                                                                                                                                                                                           |
|                | 2- Door Stuck Closed                 | Control programmed to not start if it suspects the door switch is stuck closed. Control looks for the door switch to open between cycles.<br>- Customer didn't open the door between cycles or door switch contacts stuck closed.                                                           | <ol style="list-style-type: none"> <li>1. Open and close door and then press Start/Resume key. Instruct customer to open door between cycles.</li> <li>2. Unplug dishwasher or disconnect power.</li> <li>3. Measure resistance of door switch contacts while checking mechanical operation of latch assembly.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                              |
| 6- Inlet Water | 1- Low/No Water (Mechanical Problem) | 1. No water to dishwasher.                                                                                                                                                                                                                                                                  | Verify water supply is turned on and supply line adequate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                |                                      | 2. Bowls or pots loaded or flipped and captured wash water.                                                                                                                                                                                                                                 | Instruct customer on loading. Refer to Use and Care Guide.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                |                                      | 3. Drain loop detached from tub and/ or improper drain connection.                                                                                                                                                                                                                          | <p>Check for water siphoning out of unit:</p> <ol style="list-style-type: none"> <li>1. Allow dishwasher to complete normal fill.</li> <li>2. Drain for 5–10 seconds by pressing CANCEL/DRAIN.</li> <li>3. Open door and confirm water does not siphon out of unit. If it does, confirm drain loop is attached to side of dishwasher and drain hose is connected to a drain at least 20" (50.8 cm) off the floor.</li> </ol>                                                                                                                                                                                                                                                                                           |
|                |                                      | 4. Water leaking from dishwasher.                                                                                                                                                                                                                                                           | Check for leaks under dishwasher.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



| Function Code          | Problem Code                                 | Causes                                                                       | What to Check                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|----------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6- Inlet Water (cont.) | 1- Low/No Water (Mechanical Problem) (cont.) | 5. Fill valve or water line plugged with debris.                             | Turn off water supply to dishwasher, disconnect water line to inlet valve, and inspect/clean the inlet screen of fill valve, and reconnect water.                                                                                                                                                                                                                                 |
|                        |                                              | 6. Overfill switch stuck in "Overfill" position and/or dishwasher not level. | Check other error codes to see if 6-4 also occurred. See 6-4 Error Code.                                                                                                                                                                                                                                                                                                          |
|                        |                                              | 7. Fill valve electrical problem.                                            | Check other error codes to see if 6-2 also occurred. See 6-2 Error Code.                                                                                                                                                                                                                                                                                                          |
|                        | 2- Fill Valve Electrical Problem             | 1. Loose connection in Fill Valve circuit and/or open Fill Valve Solenoid.   | Unplug dishwasher or disconnect power and check resistances of fill valve solenoid and all connections in the Fill circuit with meter.<br>-Fix/replace open connection/part.                                                                                                                                                                                                      |
|                        |                                              | 2. Open fuse on control to fill valve.                                       | Refer to "Fuse Service Check" in "Meter Check of Loads and Fuses" section.                                                                                                                                                                                                                                                                                                        |
|                        |                                              | 3. Fill Valve Drive circuit on the control.                                  | Unplug dishwasher or disconnect power and replace control.                                                                                                                                                                                                                                                                                                                        |
|                        | 3- Suds/Air in Pump                          | 1. Too many suds.                                                            | 1. Allow unit to fill and wash for 1 minute. Open door and check for excessive sudsing.<br>2. Confirm using proper dishwasher detergent, not hand detergent.<br>3. Check for excessive rinse aid leakage.                                                                                                                                                                         |
|                        |                                              | 2. Bowls or pots loaded or flipped and captured wash water.                  | Instruct customer on loading. Refer to Use and Care Guide.                                                                                                                                                                                                                                                                                                                        |
|                        |                                              | 3. Water leaking from dishwasher.                                            | Check for leaks under dishwasher.                                                                                                                                                                                                                                                                                                                                                 |
|                        | 4- Float Switch Open                         | 1. Overfill switch stuck in "Overfill" position and/or dishwasher not level. | Remove any items stuck under the float. Verify that the float moves freely and hear "click" of the switch contacts. Check/adjust levelness of the dishwasher.                                                                                                                                                                                                                     |
|                        |                                              | 2. Drain hose check valve not sealing.                                       | Water backs into dishwasher after draining and elevates water level.<br>1. Disconnect drain hose at plumbing connection.<br>2. Elevate hose above dishwasher and fill with water. If water flows into dishwasher, replace entire drain loop (install as high as possible and attach to underside of countertop if possible).                                                      |
|                        |                                              | 3. Fill valve triac on control shorted.                                      | If still filling while door is open, fill valve is mechanically stuck open (see below). If not filling with the door open, check operation in Service Diagnostics Test cycle. Advance cycle until detergent dispenser opens. Fill valve should be off. Listen to see if dishwasher is still filling. If still filling, unplug dishwasher or disconnect power and replace control. |
|                        |                                              | 4. Fill valve mechanically stuck open.                                       | Confirm dishwasher fills while the door is open. If yes, unplug dishwasher or disconnect power, turn off water to dishwasher, replace fill valve, and turn water back on.                                                                                                                                                                                                         |
|                        |                                              | 5. Too many suds.                                                            | 1. Allow unit to fill and wash for 1 minute. Open door and check for excessive sudsing.<br>2. Confirm using proper dishwasher detergent, not hand detergent.<br>3. Disconnect power and replace dispenser if excessive rinse aid leakage is seen.                                                                                                                                 |
|                        |                                              | 6. Open fuse F1 to fill valve and other triac loads.                         | Refer to "Fuse Service Check" in "Meter Check of Loads and Fuses" section.                                                                                                                                                                                                                                                                                                        |

| Function Code | Problem Code                      | Causes                                                                                                      | What to Check                                                                                                                                                                                                                                                            |
|---------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7- Heating    | 1- No Heat                        | 1. Control programmed to disable heater but continue running cycles, if it detects a water heating problem. | Running diagnostics clears the control and allows the heater to turn on again. Water heating problem must be corrected or the control will disable the heater again. See Heater circuit problem below.                                                                   |
|               |                                   | 2. Heater circuit problem:<br>- Open in heater.<br>- Open connection or component in Heater circuit.        | 1. Check operation of heater in Service Diagnostics cycle.<br>2. Unplug dishwasher or disconnect power.<br>3. Measure resistance of heater and all components and connections in Water Heating circuit/Heat Dry circuit. Fix /replace open connection/part.              |
|               |                                   | 3. Heater Drive circuit on the control.                                                                     | Unplug dishwasher or disconnect power and replace control.                                                                                                                                                                                                               |
|               | 2- Heater Stuck On                | Heater Drive circuit on the control.                                                                        | 1. Unplug dishwasher or disconnect power and replace control.<br>2. Inspect Heater and connections for overheating/shorting. If evidence of overheating or shorts exists, replace.                                                                                       |
|               |                                   |                                                                                                             |                                                                                                                                                                                                                                                                          |
| 8- Draining   | 2- Drain Motor Electrical Problem | 1. Loose connection in Drain Motor circuit and/or open Drain Motor winding.                                 | 1. Check operation of drain motor during diagnostics.<br>2. Unplug dishwasher or disconnect power.<br>3. Check resistances of drain motor winding and all connections in the Drain circuit.<br>- If high resistance, check/fix loose connections or replace drain motor. |
|               |                                   | 2. Debris stuck in drain motor impeller.                                                                    | 1. Unplug dishwasher or disconnect power.<br>2. Remove drain motor and dislodge debris from impeller.                                                                                                                                                                    |
|               |                                   | 3. Open fuse on control to Drain Motor.                                                                     | Refer to "Fuse Service Check" in "Meter Check of Loads and Fuses" section.                                                                                                                                                                                               |
|               |                                   | 4. Drain Motor Drive circuit on the control.                                                                | Unplug dishwasher or disconnect power and replace control.                                                                                                                                                                                                               |
|               | 3- Drain Stuck On                 | Drain Motor Drive circuit on the control.                                                                   | 1. Unplug dishwasher or disconnect power and replace control.<br>2. Inspect Drain Motor and connections for overheating/shorting. If evidence of overheating/shorting exists, replace.                                                                                   |
| 10- Other     | 1- Dispenser Electrical Problem   | 1. Loose connection in Dispenser circuit and/or open Dispenser solenoid.                                    | Unplug dishwasher or disconnect power and check resistances of Dispenser solenoid and all connections in the Dispenser circuit. Fix/replace open connection/part.                                                                                                        |
|               |                                   | 2. Open fuse on control to dispenser.                                                                       | Refer to "Fuse Service Check" in "Meter Check of Loads and Fuses" section.                                                                                                                                                                                               |
|               |                                   | 3. Dispenser Drive circuit on the control.                                                                  | Unplug dishwasher or disconnect power and replace control.                                                                                                                                                                                                               |
|               | 3- Drying Fan Error               | 1. Loose connection in fan circuit and/or open fan motor.                                                   | Unplug dishwasher or disconnect power and check resistances of fan motor and all connections in the fan circuit. Fix/replace open connections or fan.                                                                                                                    |
|               |                                   | 2. Fan drive circuit on the control.                                                                        | Unplug dishwasher or disconnect power and replace control.                                                                                                                                                                                                               |

# Troubleshooting Guide

## NOTES:

- For resistance checks, refer to “Dishwasher Strip Circuits” section.
- For checking operation with diagnostics, refer to “Service Diagnostics Cycle” section.

| Customer Description                                                                                                    | Potential Causes                                                                                                                          | Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Related Error Code |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Clean LED Flashes</b>                                                                                                | Control programmed with self diagnostics.                                                                                                 | Read error code from the dishwasher and refer to “Service Error Codes Table.”<br>Run Service Diagnostics Test cycle to read full history of error codes.                                                                                                                                                                                                                                                                                                                                                                             |                    |
| <b>Won't Run or Power Up (“Dead” Keypad/ Console)</b><br>- No operation<br>- No keypad response<br>- No LEDs or display | 1. No power to unit or bad connection.                                                                                                    | Check fuses, circuit breakers, and junction box connections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                    |
|                                                                                                                         | 2. Loose connections in dishwasher power-up circuit or between keypad(s) and control.                                                     | 1. Unplug dishwasher or disconnect power.<br>2. Check continuity of power connections to control.                                                                                                                                                                                                                                                                                                                                                                                                                                    |                    |
|                                                                                                                         | 3. Control detected door switch problem.                                                                                                  | 1. Unplug dishwasher or disconnect power.<br>2. Refer to “Service Error Codes Table.”                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>5-1</b>         |
|                                                                                                                         | 4. User interface or control.                                                                                                             | 1. Unplug dishwasher or disconnect power. Disassemble door and inspect control power connector (P4) and adjacent PC board for damage. Replace as needed.<br>2. Refer to “Service Error Codes Table” for stuck key (2-1). Run the diagnostic check, item (1).<br>- If drain motor turns on, control is OK. Replace the UI.<br>- If drain motor does not turn on, replace control.<br>3. Inspect UI cable for loose or damaged wiring. Replace as needed.<br>4. Inspect UI assembly for damage or contamination. Replace UI as needed. |                    |
| <b>Won't Run and LED for Start/ Resume Key is Blinking Slowly</b>                                                       | 1. By design, if the door is opened or power is interrupted during a cycle, the user must press the Start/Resume key to resume operation. | Instruct customer. Refer to Use and Care Guide.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                    |
|                                                                                                                         | 2. Start/Resume key not responding.                                                                                                       | See “One or More Keys Won't Respond.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                    |
|                                                                                                                         | 3. Control detected door switch problem.                                                                                                  | 1. Unplug dishwasher or disconnect power.<br>2. Refer to “Service Error Codes Table.”                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>5-1</b>         |
| <b>Won't Run and All LEDs On</b>                                                                                        | Software/ hardware incompatibility problem with control.                                                                                  | Refer to “Service Error Codes Table.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>1-2</b>         |
| <b>Won't Start and Start/ Resume Key LED Flashes 3 Times When Start/Resume Key is Pressed</b>                           | Control looks for door to open between cycles:<br>- Customer didn't open door between cycles.<br>- Door switch contacts stuck closed.     | Refer to “Service Error Codes Table.”                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>5-2</b>         |

| Customer Description                                                                | Potential Causes                                                                                                                            | Check                                                                                                                                                                                                                                                                                   | Related Error Code |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Won't Accept Key Presses and Control Lock LED On</b>                             | Control Lockout feature accidentally turned on by customer.                                                                                 | Instruct customer. Press and hold Control Lock key or key with Control Lock symbol next to it for 5 seconds to turn Control Lock feature off or on.                                                                                                                                     |                    |
| <b>One or More Keys Won't Respond or Unusual LED/ Display/Key Behavior</b>          | 1. Stuck key or short circuit(s) in keypad or in control's input lines that read the keys.                                                  | Refer to “Service Error Codes Table.”                                                                                                                                                                                                                                                   | <b>2-1</b>         |
|                                                                                     | 2. Loose connections between keypad and control and/or bent or contaminated connector pins.                                                 | 1. Unplug dishwasher or disconnect power.<br>2. Inspect connections in user interface circuits. Reconnect loose connections. Replace part(s) if pins are damaged or contaminated.                                                                                                       | <b>2-2</b>         |
|                                                                                     | 3. Excessive condensation on user interface parts due to vent and/or fan problem.                                                           | Check error history for 10-3 fan error. Refer to “Service Error Codes Table.”<br>Verify presence of vent current, if model has a vent wax motor. Refer to “Leaks or Drips on Cabinet or Floor” in the “Troubleshooting Guide” section.                                                  | <b>10-3</b>        |
|                                                                                     | 4. Defective user interface.                                                                                                                | 1. Unplug dishwasher or disconnect power.<br>2. Replace user interface console assembly.                                                                                                                                                                                                |                    |
| <b>Dishwasher Beeps Constantly (for models with Beepers)</b>                        | 1. User opened door during cycle and closed door without pressing Start/Resume to resume cycle.                                             | Instruct customer. Dishwasher control is designed to beep if dishwasher is in “Cycle Interrupt” mode with door latched. Control will stop beeping when door is opened and/or Start/Resume key is pressed to resume cycle.                                                               |                    |
|                                                                                     | 2. Normal beeper operation is excessive to customer.                                                                                        | Instruct customer how to turn beeper off and on. Press and hold HI TEMP for 3 seconds (tone sounds).                                                                                                                                                                                    |                    |
| <b>LEDs and/or Displays Run for Short Time (but No Loads Running) and Shuts Off</b> | 1. Unit is in Sales Demo mode.                                                                                                              | Check operation of Cancel key. If there is no Cancel LED response to multiple Cancel key presses, the control is likely in Sales Demo mode. Run Service Diagnostics cycle to clear Demo mode.                                                                                           |                    |
|                                                                                     | 2. Open F1 (triac load fuse) on control disabled loads.                                                                                     | Refer to “Fuse Service Check” in “Meter Check of Loads” section.                                                                                                                                                                                                                        |                    |
| <b>Long Cycles and/or Stuck in Certain Part of Cycle</b>                            | 1. As part of normal operation, the dishwasher pauses 2 or 3 times during the cycle for thermal holds and advances once temperature is met. | Instruct customer. Explain thermal holds and how the cycle pauses when they occur.                                                                                                                                                                                                      |                    |
|                                                                                     | 2. OWI soil sensor picking high soil cycle too often.                                                                                       | 1. Run Service Diagnostics cycle to check if OWI is showing high soil with clear water.<br>2. Check lens surface. Clean if needed.<br>3. Unplug dishwasher or disconnect power.<br>4. Replace OWI and run Diagnostics after installing new OWI to force calibration on next wash cycle. |                    |
|                                                                                     | 3. Suds/air in pump requires repeated wash periods.                                                                                         | Refer to “Service Error Codes Table.”                                                                                                                                                                                                                                                   | <b>6-3</b>         |

| Customer Description                                                                                                                                                                          | Potential Causes                                                                                     | Check                                                                                                                                                                                                                                                                                                                                                     | Related Error Code       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Long Cycles and/or Stuck in Certain Part of Cycle (cont.)</b>                                                                                                                              | 4. A water heating problem could cause long cycles but will typically cause a "water heating fault." | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>7-1</b>               |
|                                                                                                                                                                                               | 5. Heater takes a long time to heat water with low voltage.                                          | Check for at least 100 VAC at power source.                                                                                                                                                                                                                                                                                                               |                          |
|                                                                                                                                                                                               | 6. Incoming water under 84°F (29°C).                                                                 | 1. Be sure dishwasher is connected to the hot water supply.<br>2. Confirm temperature at sink (recommend 120°F [49°C]). Instruct customer to run water at sink before running dishwasher.<br>3. Unplug dishwasher or disconnect power and check all connections and measure resistance in Temperature Sensing circuit. Replace OWI if resistance is high. |                          |
|                                                                                                                                                                                               | 7. OWI or NTC sensor problem.                                                                        | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>3-1</b><br><b>3-3</b> |
| <b>Can Start a Cycle, but Cycle Does Not Complete (and Clean LED or Completed May Blink)</b>                                                                                                  | 1. Unit is in Sales Demo mode.                                                                       | Run Service Diagnostics Cycle to clear Demo mode.                                                                                                                                                                                                                                                                                                         |                          |
| <b>Will Not Drain or Excess Water Left in Unit</b><br><b>NOTE:</b> Check error history. If no error codes for electrical problems, problem is mechanical. Do not replace control.             | 1. Drain loop check valve not sealing.                                                               | 1. Disconnect drain hose at plumbing connection.<br>2. Elevate hose above dishwasher and fill with water. If water flows into dishwasher, replace entire drain loop (install as high as possible).                                                                                                                                                        |                          |
|                                                                                                                                                                                               | 2. Customer misunderstands water level after drain.                                                  | Instruct customer. Sump will normally have about 1" (2.4 cm) of water remaining after cycle.                                                                                                                                                                                                                                                              |                          |
|                                                                                                                                                                                               | 3. Draining problem                                                                                  | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>8-2</b>               |
|                                                                                                                                                                                               | 4. Obstructed drain hose or path.                                                                    | 1. Unplug dishwasher or disconnect power.<br>2. Check for blockages from Sump Check Valve to customer's plumbing. Potential items: plugged garbage disposal or plug not knocked out, drain loop Check Valve stuck, and/or plugged hoses.                                                                                                                  |                          |
|                                                                                                                                                                                               | 5. Drain Pump impeller fractured.                                                                    | 1. Unplug dishwasher or disconnect power.<br>2. Remove Drain Pump and check impeller (normally there is some uneven resistance). If it is stripped, replace Drain Pump.                                                                                                                                                                                   |                          |
| <b>Detergent Not Dispensing or Detergent Left in Dispenser</b><br><b>NOTE:</b> Check error history. If no error codes for electrical problems, problem is mechanical. Do not replace control. | 1. Item in lower rack blocked lid or blocked spray of water to dispenser.                            | Instruct customer on proper dish loading.                                                                                                                                                                                                                                                                                                                 |                          |
|                                                                                                                                                                                               | 2. Mechanical binding of dispenser lid.                                                              | 1. Unplug dishwasher or disconnect power.<br>2. Check/replace dispenser.                                                                                                                                                                                                                                                                                  |                          |
|                                                                                                                                                                                               | 3. Lid latch binding due to excess detergent in mechanism.                                           | Instruct customer on proper dispenser filling.                                                                                                                                                                                                                                                                                                            |                          |
|                                                                                                                                                                                               | 4. Dispenser electrical problem.                                                                     | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>10-1</b>              |

| Customer Description                          | Potential Causes                                                        | Check                                                                                                                                                                                                                                                                    | Related Error Code       |
|-----------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Poor Wash</b>                              | 1. Cycle selection by customer not appropriate for dish load.           | Instruct customer on cycle selection. Recommend High Temp option for wash performance boost.                                                                                                                                                                             |                          |
|                                               | 2. Spray arms plugged or not rotating.                                  | 1. Check arm rotation. If arms are blocked by dish item, instruct customer. Also check for correct upper spray arm alignment with docking station located on feed tube back wall.<br>2. Check nozzles. If plugged, clean nozzles and confirm filters installed properly. |                          |
|                                               | 3. Poor wash due to draining, dispensing, and/or temperature problem.   | See "Will Not Drain or Excess Water Left in Unit" or "Detergent not Dispensing or Detergent Left in Dispenser," or details on temperature sensing in "Long Cycles and/or Stuck in Certain Part of Cycle."                                                                |                          |
|                                               | 4. Soil sensor problem.                                                 | Refer to "Service Error Codes Table."<br>NOTE: Even if no error code recorded, confirm OWI passes all OWI checks in Service Diagnostics cycle, and see checks for Error 3-3.                                                                                             | <b>3-2</b><br><b>3-3</b> |
|                                               | 5. Heating problem.                                                     | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                    | <b>7-1</b>               |
| <b>Film or Spots on Glasses and/or Dishes</b> | 1. Customer not using rinse aid or dispenser empty.                     | Check rinse aid gauge level on dispenser. Instruct customer how to fill and monitor add/use rinse aid.                                                                                                                                                                   |                          |
|                                               | 2. Rinse aid dispenser problem.                                         | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                    | <b>10-1</b>              |
|                                               | 3. Hard water leaving film on dishes.                                   | Check water hardness. If hard, instruct customer to use dishwasher cleaner according to packaging instructions. Also recommend the 1 HR Wash cycle.                                                                                                                      |                          |
|                                               | 4. Detergent carryover or over-sudsing.                                 | Check water hardness. If below 10 grains, then instruct customer to use less detergent and recommend the 1 HR Wash cycle.                                                                                                                                                | <b>6-3</b>               |
|                                               | 5. Etching of glass from too much detergent at too high of temperature. | Check water hardness. If below 10 grains, then instruct customer to use less detergent and recommend the 1 HR Wash cycle.                                                                                                                                                |                          |
|                                               | 6. Drain loop check valve not sealing.                                  | 1. Disconnect drain hose at plumbing connection.<br>2. Elevate hose above dishwasher and fill with water. If water flows into dishwasher, replace entire drain loop (install as high as possible).                                                                       |                          |
| <b>Poor Drying</b>                            | 1. Customer not using rinse aid and/or dispenser empty.                 | Check rinse aid gauge level on dispenser. Instruct customer how to fill and monitor and add/use rinse aid.                                                                                                                                                               |                          |
|                                               | 2. Customer not using Heated Dry option.                                | Recommend use of Heated Dry to customer.                                                                                                                                                                                                                                 |                          |
|                                               | 3. Heating problems.                                                    | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                    | <b>7-1</b>               |
|                                               | 4. Rinse aid dispenser problem.                                         | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                    | <b>10-1</b>              |
|                                               | 5. Fan problem (on models with fan).                                    | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                    | <b>10-3</b>              |

| Customer Description                                                                                                              | Potential Causes                                                                                                                    | Check                                                                                                                                                                                                                                                                                                                                                     | Related Error Code                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Sanitized LED Blinks or Incomplete Sanitization Message at End of Cycle (Control Could Not Confirm Sanitization Achieved).</b> | 1. Door opened during final rinse or dry.                                                                                           | Instruct customer.                                                                                                                                                                                                                                                                                                                                        |                                                      |
|                                                                                                                                   | 2. Incoming water under 84°F (29°C).                                                                                                | 1. Be sure dishwasher is connected to the hot water supply.<br>2. Confirm temperature at sink (recommend 120°F [49°C]). Instruct customer to run water at sink before running dishwasher.<br>3. Unplug dishwasher or disconnect power and check all connections and measure resistance in Temperature Sensing circuit. Replace OWI if resistance is high. |                                                      |
|                                                                                                                                   | 3. Line voltage too low to heat fast enough.                                                                                        | Check power source. confirm at least 100 VAC.                                                                                                                                                                                                                                                                                                             |                                                      |
|                                                                                                                                   | 4. Heating problem.                                                                                                                 | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>7-1</b>                                           |
|                                                                                                                                   | 5. Intermittent door switch/latch connection.                                                                                       | See same checks as for 5-1 error. Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                   |                                                      |
|                                                                                                                                   | 6. Thermistor/OWI problem.                                                                                                          | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>3-1</b><br><b>3-2</b>                             |
| <b>Melted Dishware and/or Spray Arm and/or Dishwasher Always Hot</b>                                                              | 7. Air pressure surges in dishwasher due to washing with high suds causes brief opening of door switch contacts during final rinse. | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>6-3</b>                                           |
|                                                                                                                                   | 1. Customer uses non-dishwasher-safe dishes or loads directly over heater.                                                          | Instruct customer.                                                                                                                                                                                                                                                                                                                                        |                                                      |
|                                                                                                                                   | 2. Water heater displaced from mounting clip and/or pulled off center.                                                              | Inspect heater. Adjust back into position if needed.                                                                                                                                                                                                                                                                                                      |                                                      |
|                                                                                                                                   | 3. Water heating problem - heater stuck on.                                                                                         | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>7-2</b>                                           |
| <b>Noisy Operation</b>                                                                                                            | 4. Temperature sensing problem.                                                                                                     | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>3-1</b>                                           |
|                                                                                                                                   | 1. Spray arm stalled or blocked and spraying on the door.                                                                           | - Instruct customer if blocked.<br>- Check spray arm rotation and inspect for plugged nozzles. If plugged, clean nozzles and confirm filters are installed properly.                                                                                                                                                                                      |                                                      |
|                                                                                                                                   | 2. No or low water.                                                                                                                 | Refer to "Service Error Codes Table."                                                                                                                                                                                                                                                                                                                     | <b>6-1</b><br><b>6-2</b><br><b>6-3</b><br><b>6-4</b> |
|                                                                                                                                   | 3. Drains too long.                                                                                                                 | 1. Long drain due to OWI sensor problem. Refer to "Service Error Codes Table" for 3-3.<br>2. Slow drain problem. See "Will Not Drain or Excess Water Left in Unit" for "Obstructed drain hose or path" and "Drain Pump impeller fractured".                                                                                                               | <b>3-3</b>                                           |

| Customer Description                      | Potential Causes                                                                                                                 | Check                                                                                                                                                                                       | Related Error Code       |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Noisy Operation (cont.)</b>            | 4. Fan runs (makes noise) after cycle completed (on models with fan).                                                            | Dishwasher is designed to keep fan running after cycle to avoid moisture buildup in dishwasher. Fan will turn off if door is opened or if any key is pressed.                               |                          |
|                                           | 5. Excessive fan noise due to faulty fan (on models with fan).                                                                   | 1. Check fan operation during Service Diagnostics Test cycle.<br>2. Unplug dishwasher or disconnect power.<br>3. Replace fan if fan does not spin freely.                                   |                          |
| <b>Leaks or Drips on Cabinet or Floor</b> | 1. Too many suds.                                                                                                                | Refer to "Service Error Codes Table."                                                                                                                                                       | <b>6-3</b><br><b>6-4</b> |
|                                           | 2. Leaking dishwasher.                                                                                                           | Check door/tub gasket and all water connections under dishwasher. If leak triggers low water error, leak is probably not from water inlet connection. Refer to "Service Error Codes Table." | <b>6-1</b><br><b>6-3</b> |
|                                           | 3. Unit not level (leaning forward) and water surges over front lip during cycle.                                                | Check error history for float error 6-4. Error 6-4 is likely to occur if unit is significantly unlevel and leaning forward. Refer to "Service Error Codes Table."                           | <b>6-4</b>               |
|                                           | 4. Air pressure surge when open door and immediately re-closing it while dishwasher is hot can force droplets out the vent duct. | Instruct customer to leave door open a few minutes before re-closing if opened door while dishwasher is hot.                                                                                |                          |
|                                           | 5. Fan problem (on models with fan.)                                                                                             | Refer to "Service Error Codes Table."                                                                                                                                                       | <b>10-3</b>              |